AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 23 · TEACHER GUIDE

Recommendation Scores

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Teach recommendation scores as declared policies with visible constraints.

Supply complete records and scripts, separate the cases and preserve no-eligible and review outcomes.

Keep one habit: Require reasons and component arithmetic before accepting a top-score recommendation.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Compute a **weight**ed **score**

Teach Week 23 as a two-session investigation using fictional data. A fictional activity chooser uses enjoyment and learning ratings from 0 to 5. Its **score** is 2 times enjoyment plus learning.

The activity must also take at most 20 minutes. Duration is a hard requirement; a high **score** cannot compensate for exceeding it. Assess the three stated outcomes through the independent task, not by copying the worked result.

By the end, I can **rank** only eligible options with a transparent **weight**ed **score**.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Python 3 editor with a fresh local practice file
- Paper trace table and the student workbook
- Fictional **input** records from this lesson; no real student identifiers

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Nearest Neighbor Reasoning. Give an example and a mistake that the idea helps you avoid.

Look for: Compute absolute distances and **rank training** neighbours

Reference example, not the only acceptable wording: Distances to A,B,C,D,E are 1,1,5,7,9 cm. A and B are equally far from 3 on the numeric feature; their labels do not enter the distance calculation.

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

Look for: Identify selection and coverage bias

Reference example, not the only acceptable wording: Afternoon or evening visitors, and people unable to attend in the morning.

Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate mean and median

Reference example, not the only acceptable wording: Sum 30 divided by 5 gives 6.

A gap of 2.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Nearest Neighbor Reasoning

Repair Bias in Data

Noise and Outliers

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Score=3×genreMatch+2×short+illustrated. Compute A=[1,0,1], B=[0,1,1], C=[1,1,0]. A reader now requires illustrations. Which book wins after applying that **constraint**?

Concept to explain

A **weight**ed **score** expresses priorities but also embeds choices.

Term	Meaning
weight	A multiplier controlling how strongly a feature or value contributes to a calculation.
score	A numerical value assigned using a specified rule; it is not automatically a probability.
rank	Place candidates in an order using a declared comparison rule.
tradeoff	A situation where improving one desired property can make another harder to achieve.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach must-have checks before a preference ranking

> Keep the20-,14- and15-minute cases distinct.

State the opening rule and full script

A fictional activity chooser uses enjoyment and learning ratings from 0 to 5. Its **score** is 2 times enjoyment plus learning. The activity must also take at most 20 minutes.

Duration is a hard requirement; a high **score** cannot compensate for exceeding it.

```
options = [{"name": "A", "fun": 4, "learning": 2, "minutes": 15},
           {"name": "B", "fun": 3, "learning": 5, "minutes": 25}]
for item in options:
    if item["minutes"] <= 20:
        score = 2 * item["fun"] + item["learning"]
        print(item["name"], score)
```

Explain inclusive eligibility and collect the changed weights

Only A is printed, with **score** 10. If B took 20 minutes it would qualify and have the higher **score** 11. The inclusive boundary matters.

Ratings and **weight**s are invented for practice and require justification before use in a real decision.

> **If enjoyment weight becomes 1 and learning weight becomes 2, find both scores before applying the same duration limit.**

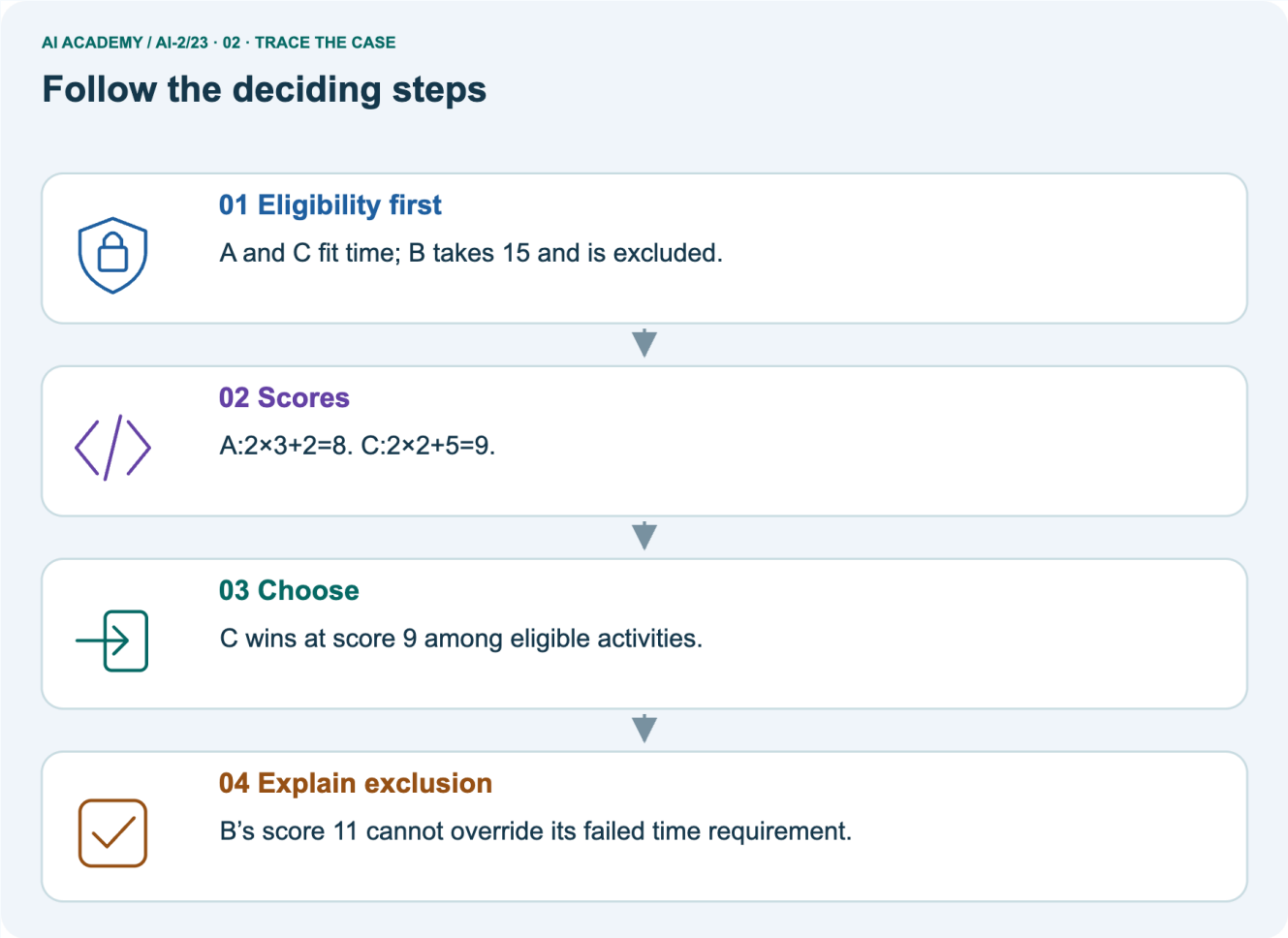
Introduce the separate14-minute case

A fictional book-activity chooser has a 14-minute limit.

Score=2×interest+practice. A=(interest3,practice2,time14); B=(4,3,15); C=(2,5,12).

Require whole nonnegative feature values; time must be at most the limit. Highest **score** wins; tie by ID.

Trace its eligible scores and limits



Follow the labelled picture with the matching explanation. Open the vector diagram

A **score** combines preferences according to explicitly chosen **weight**s; it does not erase hard **constraint**s.

Computing a high **score** for an excluded item can be useful in an audit, but it must not allow that item back into the candidate set. Store both eligibility reasons and **score** components for explanations.

Toy interest **score**s are supplied **input**s, not measured enjoyment or a universal definition of a good activity.

Change the interest weight without changing eligibility

Change: Change interest **weight** from 2 to 3, keeping time limit 14.

Prediction: A **score**s 11; C also **score**s 11; tie-by-ID chooses A.

Reason: B remains too long; changing preferences changes only **rank**ing of A and C.

Run the complete audit, tie and empty-set branch

```
options = [{"id": "A", "interest": 3, "practice": 2, "minutes": 14},
           {"id": "B", "interest": 4, "practice": 3, "minutes": 15},
           {"id": "C", "interest": 2, "practice": 5, "minutes": 12}]
limit = 14
for interest_weight in [2, 3]:
    scores = {}
    review = []
    for item in options:
        values = [item.get(key) for key in ["interest", "practice", "minutes"]]
        if not all(type(value) is int and value >= 0 for value in values):
            review.append((item["id"], "missing or invalid input"))
        elif item["minutes"] > limit:
            review.append((item["id"], "exceeds time limit"))
        else:
            scores[item["id"]] = interest_weight * item["interest"] +
item["practice"]
    print("weight", interest_weight, "scores", scores, "review", review)
    if not scores:
        print("No eligible option")
    else:
        highest = max(scores.values())
        tied = sorted(name for name in scores if scores[name] == highest)
        print("top", tied, "chosen", tied[0])
```

Elicit a justified explanation

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. **Score** = 3×genre match + 2×difficulty match + 1×length match.

2. Book A matches genre and length: 3+0+1=4.

3. Book B matches difficulty and length: 0+2+1=3.

4. A **rank**s first for these **weight**s—not as a universal best book.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Score=2 times genre match + length match. A matches both, B genre only, C length only.

Score and **rank**.

Exact explanation to model

A=3, B=2, C=1. **Weight**s encode a chosen preference, not universal reader truth.

Explain the added worked example

Key terms

Term	Meaning
Weight	A multiplier expressing the importance assigned to a feature
Score	A number calculated to compare options

Term	Meaning
Constraint	A requirement an option must meet
Tradeoff	Improving one preference while giving up another

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Recommendation Scores

The facts we will use

AI ACADEMY / AI-2/23

Time filtering and scores

	Minutes	Score	Status
A	14	8	eligible
B	15	11	excluded
C	12	9	eligible

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/23

Recommendation Scores

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Eligibility first

A and C fit time; B takes 15 and is excluded.



2. Scores

A:2 x 3+2=8. C:2 x 2+5=9.



3. Choose

C wins at score 9 among eligible activities.



4. Explain exclusion

B's score 11 cannot override its failed time requirement.

CHANGE ONE THING

Change interest weight from 2 to 3, keeping time limit 14.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Eligibility first	A and C fit time; B takes 15 and is excluded.
2. Scores	A:2×3+2=8. C:2×2+5=9.
3. Choose	C wins at score 9 among eligible activities.
4. Explain exclusion	B's score 11 cannot override its failed time requirement.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: Which durations 19, 20, 21 qualify under at most 20?

Expected reasoning and fading help

Teacher check: 19 and 20 qualify; 21 does not.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for **Weight**ed **score**s and **constraint**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: **Weight**s represent preferences, not a discovered truth. Explain who chose them and test whether small changes alter the recommendation. Avoid claiming that a larger **score** proves greater educational value.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

Why does B not become the winner when its **score** is largest?

Its time 15 exceeds the 14-minute maximum, so it is ineligible before **rank**ing.

Changed-case reasoning: A **score**s 11; C also **score**s 11; tie-by-ID chooses A. B remains too long; changing preferences changes only **rank**ing of A and C.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: **Score**=2 times genre match + length match. A matches both, B genre only, C length only.

Write the question to explain: **Score** and **rank**.

Reveal after **prediction**s: A=3, B=2, C=1. **Weight**s encode a chosen preference, not universal reader truth.

Use the earlier changed question as optional later practice. Reserve Choose independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

A **score**s 8 and B **score**s 13. B remains excluded at 25 minutes. Exit: First **score**s A 9, B 8 so A wins. Second **score**s A 6, B 10 so B wins. Explain the preference change.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

Opening A15minutes scores10; B25 is excluded despite audit score11. If B were20 it would be eligible11. Changed opening weights(1,2) give A8,B13 but B remains excluded at25. Separate14-minute lab: weight2 gives A8,C9,chooseC; weight3 gives A11,C11,tieA/C and ID choosesA; B15 remains excluded. Limit0 on a separate copy produces no eligible option. W5 eligible options give9/8 then6/10. Never call a declared weight a measured educational benefit. Reserve the15-minute independent case for W4.

Run and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.
3. Use the supplied14-minute A/B/C records and complete two-**weight** comparison. Retain eligible scores, excluded rows and reasons, all top ties and the declared ID choice. Test no eligible option on a separately labelled changed-limit copy; do not invent reader measurements.
4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

A successful run proves the conclusion: **Weight**s represent preferences, not a discovered truth. Explain who chose them and test whether small changes alter the recommendation.

Avoid claiming that a larger **score** proves greater educational value.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: limit 10, X (fun 1, learning 5, minutes 9), Y (3,2,10), Z (5,5,11). **Score**s 7, 8, 15; choose Y, exclude Z.

A convenient shortcut preserves meaning: Choosing the highest **score** and only then ignoring its duration breaks the requirement. Filter for eligibility first.

If no item qualifies, report no eligible option instead of relaxing a requirement silently.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Choose independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new eligible-choice case

> Keep this key until after the independent first attempt.

Selected W4 and checked key

Selected case: Limit is 15 minutes. A has fun 3, learning 4, minutes 15; B has 5, 1, 16; C has 2, 5, 12. Calculate **score**s with 2 * fun + learning and choose.

Scores A 10, B 11, C 9. B is excluded. A is highest among eligible options.

Judge components, selection and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give a repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 score key

An eligible activity has fun 2 and learning 4. Use 2 * fun + learning.

Score 8.

Substitute each rating into its named role: 2 * fun + learning = 2 * 2 + 4 = 4 + 4 = 8. Eligibility is supplied in this task, so this calculation **rank**s an already permitted option.

The **score** is a result of the chosen preference rule, not a measured educational benefit.

W2 boundary key

Which durations 19, 20, 21 qualify under at most 20?

19 and 20 qualify; 21 does not.

At most 20 means minutes <= 20. The value 19 is below the limit and 20 is exactly at it, so both qualify. The value 21 is above it and is excluded.

An answer that rejects 20 has changed the inclusive requirement into a strict one.

W3 eligibility key

An ineligible item **score**s 15 and an eligible item **score**s 9. Which should be offered under the stated policy?

Offer the eligible item with **score** 9. The **constraint** is applied before **rank**ing.

W5 sensitivity key

For eligible A (fun 4, learning 1) and B (fun 2, learning 4), compare **weight**s (2,1) with (1,2).

First **score**s A 9, B 8 so A wins. Second **score**s A 6, B 10 so B wins. Explain the preference change.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

A=3, B=2, C=1. **Weight**s encode a chosen preference, not universal reader truth.

Additional transfer practice question

A recommender gives 3 points for topic match and 1 for difficulty match. Book A matches topic only; B matches difficulty only. Compute **score**s and justify the **rank**ing.

Reasoned practice answer

A **score**s 3 and B **score**s 1, so A **rank**s first. The **rank**ing reflects the chosen **weight**s; different priorities could change them.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Recommendation **score**s are objective preferences of all readers.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Scores A=4, B=3, C=5. With illustrated required, C is excluded and A wins. A high **score** cannot compensate for a mandatory requirement unless the requirement has explicitly changed.

Explanations for the additional workbook examples

W1 Calculate a score

W2 Check a boundary

W3 Repair selection

W4 Choose independently

Limit is 15 minutes. A has fun 3, learning 4, minutes 15; B has 5, 1, 16; C has 2, 5, 12. Calculate **score**s with 2 * fun + learning and choose.

Scores A 10, B 11, C 9. B is excluded. A is highest among eligible options.

W5 Test sensitivity

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Publish the features, **weight**s, raw **score**s, and reason for the top result.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Publish the recommender's **constraint**s, **score** formula and tie rule; include component-level reasons in each suggestion.

Support while preserving the reasoning goal

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Use separate eligibility and **score** columns. Extension: add a documented tie rule and test tied eligible items.

Independent W4 **evidence**: **Score**s A 10, B 11, C 9. B is excluded. A is highest among eligible options. Record whether the learner needed no help, a hint or explanation.

Fresh retry: limit 10, X (fun 1, learning 5, minutes 9), Y (3,2,10), Z (5,5,11). **Score**s 7, 8, 15; choose Y, exclude Z.

Plan the delayed check

Suggested return points in this level: ai-2/24, ai-2/26, ai-2/29. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	

Record	Teacher evidence
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 24: Feedback Loops. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week24: Feedback Loops. Carry forward published score/eligibility rules and track how displayed suggestions influence the feedback later used to revise them.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Transparent Book Recommender

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Compute a **weight**ed **score**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

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>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: Which durations 19, 20, 21 qualify under at most 20?

Reasoned practice answer: 19 and 20 qualify; 21 does not.

Check every learner before discussion: Use the worked case for **Weight**ed **score**s and **constraint**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: **Weight**s represent preferences, not a discovered truth. Explain who chose them and test whether small changes alter the recommendation. Avoid claiming that a larger **score** proves greater educational value.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change interest **weight** from 2 to 3, keeping time limit 14. Require a **prediction** and a reason before comparing. Expected result: A **score**s 11; C also **score**s 11; tie-by-ID chooses A.

Reason: B remains too long; changing preferences changes only **rank**ing of A and C.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery **score**s. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

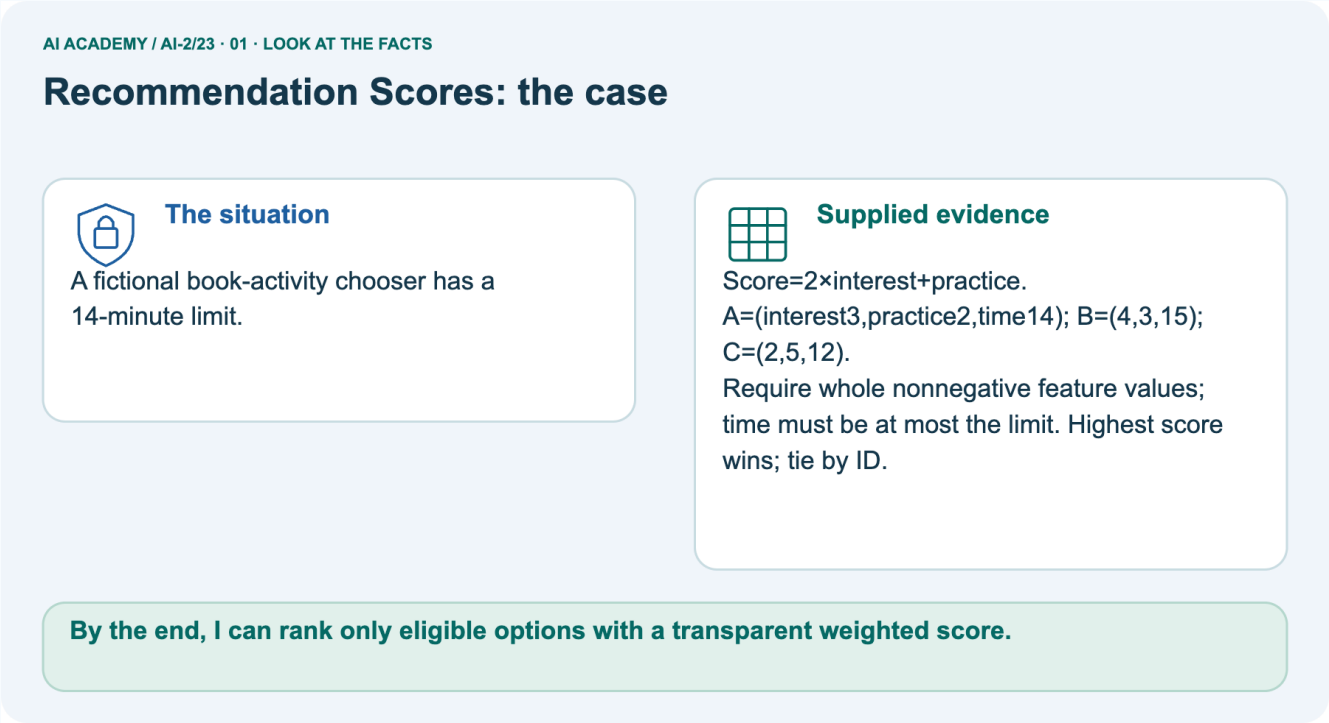
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Recommendation Scores: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

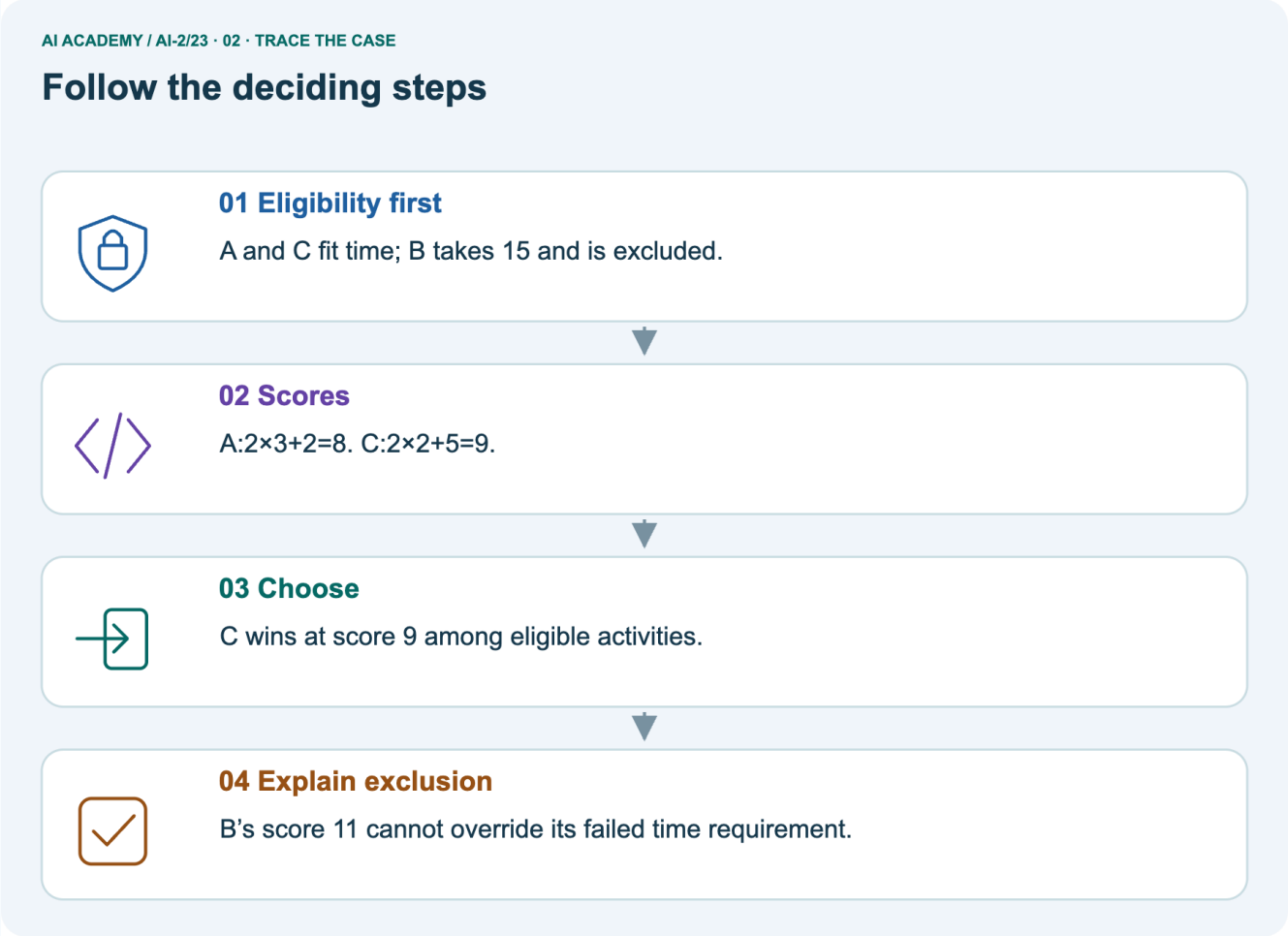
A fictional book-activity chooser has a 14-minute limit. Supplied facts: **Score**=2×interest+practice. A=(interest3,practice2,time14); B=(4,3,15); C=(2,5,12).

Require whole nonnegative feature values; time must be at most the limit. Highest **score** wins; tie by ID.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

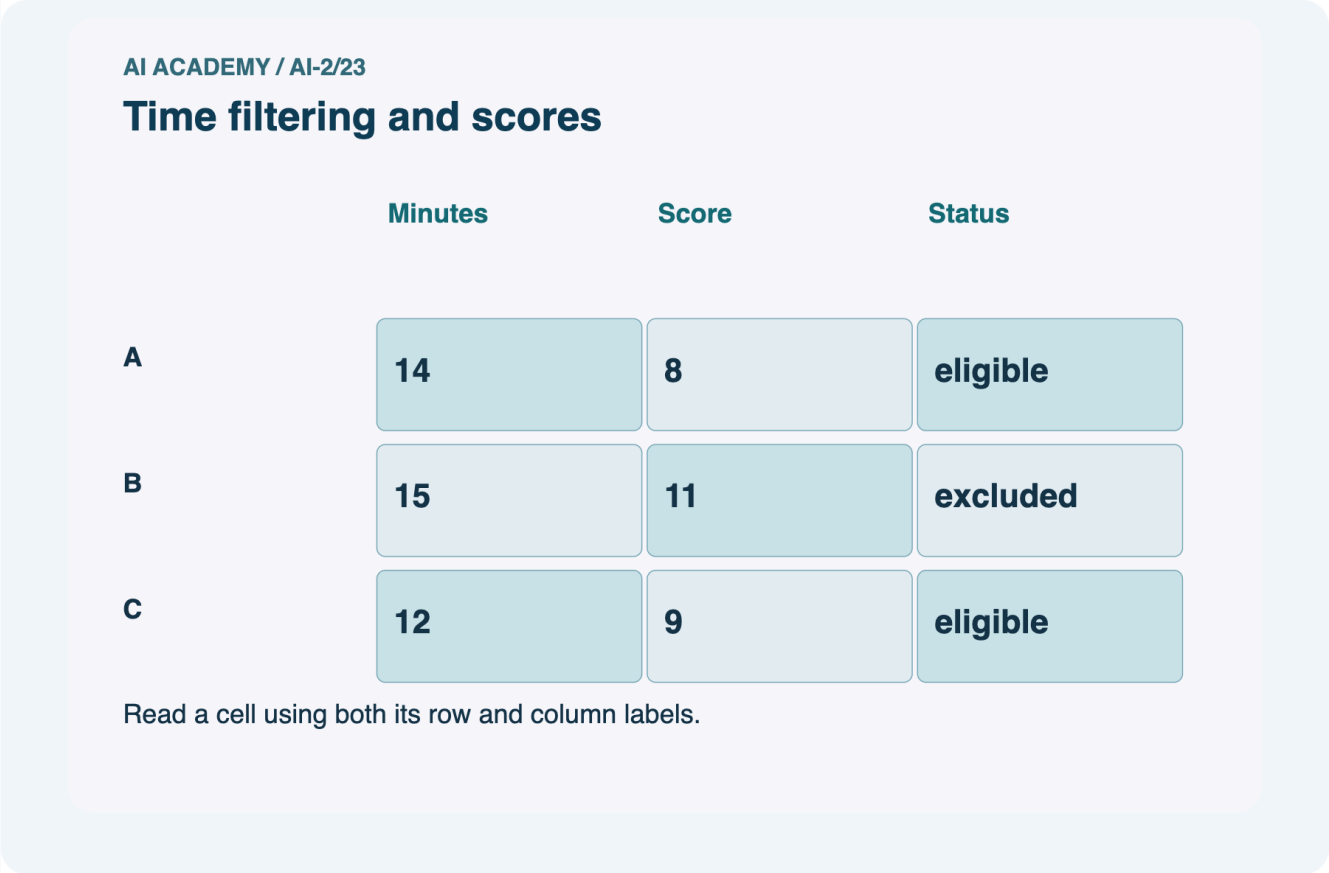
Eligibility first: A and C fit time; B takes 15 and is excluded. → **Score**s: A:2×3+2=8.

C:2×2+5=9. → Choose: C wins at **score** 9 among eligible activities. → Explain exclusion: B's **score** 11 cannot override its failed time requirement.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Time filtering and scores



Follow the labelled picture with the matching explanation. Open the vector diagram

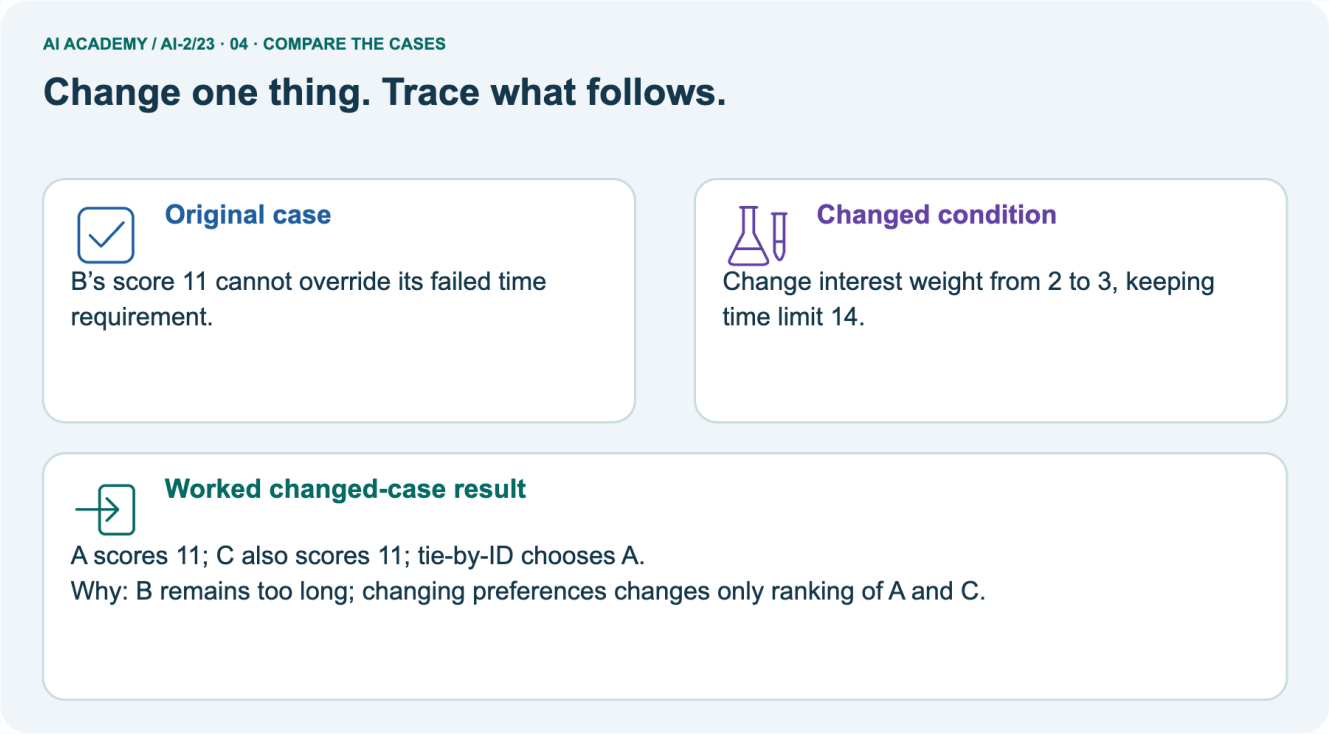
Time filtering and **score**s. A: Minutes = 14, **Score** = 8, Status = eligible. B: Minutes = 15, **Score** = 11, Status = excluded. C: Minutes = 12, **Score** = 9, Status = eligible.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B's **score** 11 cannot override its failed time requirement. Changed condition: Change interest **weight** from 2 to 3, keeping time limit 14. Expected result: A **score**s 11; C also **score**s 11; tie-by-ID chooses A.

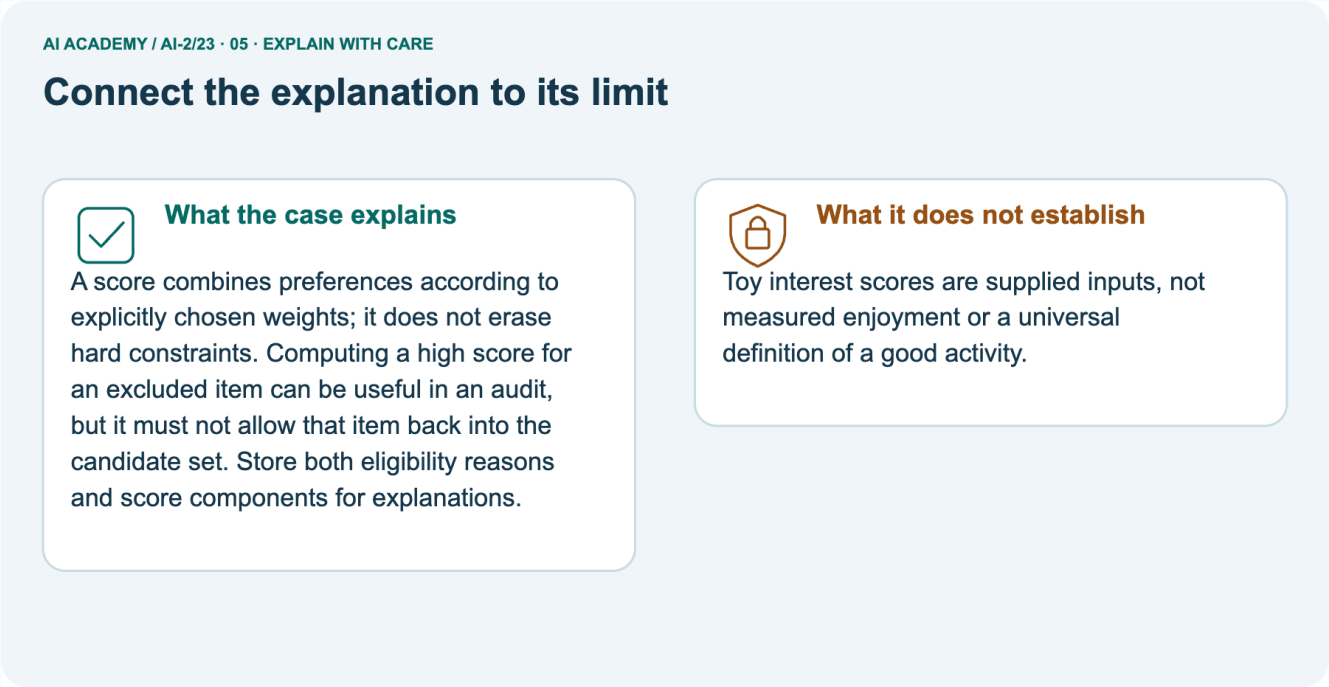
Reason: B remains too long; changing preferences changes only **rank**ing of A and C.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: B remains too long; changing preferences changes only **rank**ing of A and C.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **score** combines preferences according to explicitly chosen **weight**s; it does not erase hard **constraint**s.

Computing a high **score** for an excluded item can be useful in an audit, but it must not allow that item back into the candidate set. Store both eligibility reasons and **score** components for explanations.

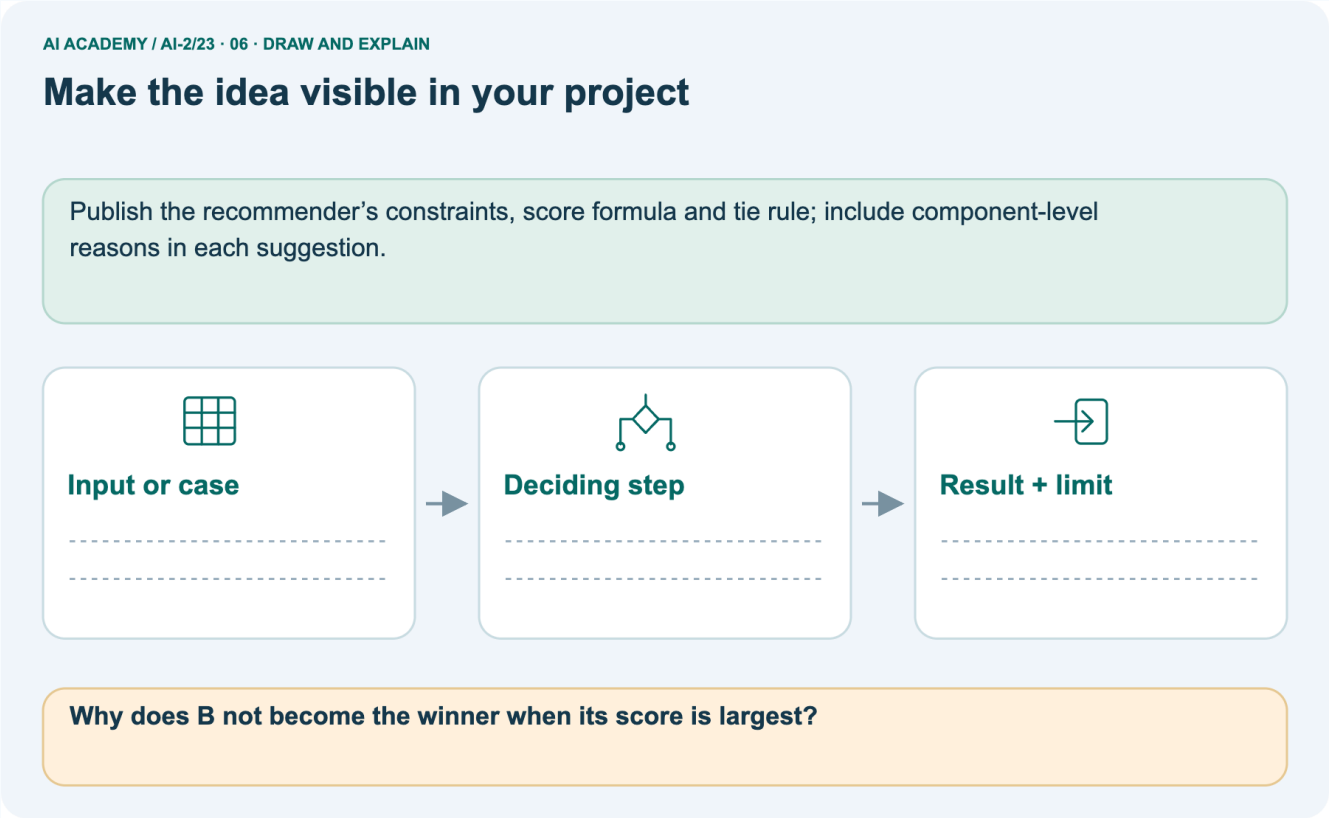
Limit: Toy interest **score**s are supplied **input**s, not measured enjoyment or a universal definition of a good activity.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Toy interest **score**s are supplied **input**s, not measured enjoyment or a universal definition of a good activity.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Publish the recommender's **constraint**s, **score** formula and tie rule; include component-level reasons in each suggestion. Draw your **input**, deciding step, result and limit.

Supported question: Why does B not become the winner when its **score** is largest?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Its time 15 exceeds the 14-minute maximum, so it is ineligible before **rank**ing.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Recommendation Scores: the case

02. Follow the deciding steps

03. Time filtering and scores

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.